MERN Backend Assessment - Answer Key

API Consumption, JWT Authentication, and Data Handling

SET A – Hospital (Base Route: /patients)

Q1: Highest Total Hospital Bill

GET /patients/highest-bill

Output

```
{
  "id": 250,
  "name": "Girish",
  "department": "Neurology",
  "totalBill": 10700
}
```

Q2: Get Patient by ID

GET /patients/:id

Example

GET /patients/210

Output

```
{
  "id": 210,
  "name": "Lakshmi",
  "gender": "Female",
  "age": 49,
  "department": "Cardiology",
  "doctor": "Dr. Ravi",
  "diagnosis": "Arrhythmia",
  "bill": {
    "consultation": 2100,
    "medicine": 4000,
    "lab": 2000
  },
  "daysAdmitted": 6,
  "status": "Admitted"
}
```

Q3: Filter by Department

GET /patients/filter?department=VALUE

Example

GET /patients/filter?department=Cardiology

Output

```
[
  {
    "id": 201,
    "name": "Arun",
    "gender": "Male",
    "age": 34,
    "department": "Cardiology",
    "doctor": "Dr. Ravi",
    "diagnosis": "Hypertension",
    "bill": {
      "consultation": 1500,
      "medicine": 3200,
      "lab": 1800
    },
    "daysAdmitted": 3,
    "status": "Discharged"
  },
Etc., (10 patients)
```

Q4: Longest Admitted Patient

GET /patients/longest-stay

Output

```
{
  "id": 250,
  "name": "Girish",
  "department": "Neurology",
  "daysAdmitted": 9
}
```

Q5: Hospital Admission Analysis

GET /patients/admission/summary

Output

```
{
  "admittedCount": 24,
  "dischargedCount": 26,
  "averageAge": 38.2
}
```

SET B – E-Commerce (Base Route: /products)

Q1: Highest Rated Product

GET /products/top-rated

Output:

```
{
  "id": 28,
  "name": "Data Science Handbook",
  "category": "Books",
  "rating": 4.9
}
```

Q2. Get Product by ID

GET /products/:id

Example:

GET /products/8

Output:

```
{
  "id": 8,
  "name": "Mechanical Keyboard",
  "category": "Electronics",
  "brand": "KeyPro",
  "price": 4500,
  "stock": 22,
  "rating": 4.4,
  "discount": 6
}
```

Q3: Filter by Category

GET /products/filter?category=VALUE

Example:

GET /products/filter?category=Electronics

Output:

```
[
  {
    "id": 1,
    "name": "Laptop Pro 15",
    "category": "Electronics",
    "brand": "TechX",
    "price": 75000,
```

```
    "stock": 12,
    "rating": 4.6,
    "discount": 10
  },
Etc…. (15 products)
```

Q4. Discounted Price Calculation

GET /products/discount/:id

Example:

GET /products/discount/10

Output:

```
{
  "id": 10,
  "name": "4K Monitor",
  "originalPrice": 28000,
  "discountPercentage": 9,
  "finalPrice": 25480
}
```

Q5. Inventory Summary

GET /products/inventory/summary

```
Output:
{
  "totalStock": 1861,
  "totalInventoryValue": 13345500
}
```

SET C – HR (Base Route: /employees)

(As per SET C paper)

Q1. Highest Salary

GET /employees/highest-salary

Output:

```
{
  "id": 140,
  "name": "Girish",
  "department": "Engineering",
  "role": "Cloud Architect",
  "totalSalary": 125000
}
```

Q2. Get Employee by ID

GET /employees/:id

Example:

GET /employees/112

Output:

```
{
  "id": 112,
  "name": "Rahul",
  "department": "Engineering",
  "role": "Backend Developer",
  "salary": {
    "basic": 55000,
    "bonus": 9000
  },
  "experience": 5,
  "performanceRating": 4.5,
  "status": "Active"
}
```

Q3. Filter by Department

GET /employees/filter?department=VALUE

Example:

GET /employees/filter?department=Engineering

Output:

```
[
  {
    "id": 101,
    "name": "Ravi",
    "department": "Engineering",
    "role": "Backend Developer"
  },
Etc.. employees(23)
```

Q4. Senior Employees by Experience

GET /employees/senior/:years

Example:

GET /employees/senior/8

Output:

```
[
  {
    "id": 140,
    "name": "Girish",
    "experience": 12,
    "department": "Engineering"
  }
]
```

Q5. Experience & Performance Summary

GET /employees/analysis/summary

Output:

```
{
  "averageExperience": 5.86,
  "highPerformersCount": 21,
  "activeEmployeesCount": 47
}
```

SET D – Orders (Base Route: /orders)

Q1. Order Total

GET /orders/total/:orderId

Example:

GET /orders/total/5001

Output:

```
{
  "orderId": 5001,
  "customerName": "Arun",
  "orderTotal": 75500
}
```

Q2. Orders by Status

GET /orders/status/:status

Example:

GET /orders/status/delivered

Output:

```
[
  {
    "orderId": 5001,
    "customerName": "Arun",
    "status": "Delivered"
  }
Etc.. orders(13)
```

Q3. Filter by Customer Type

GET /orders/filter?customerType=VALUE

Example:

GET /orders/filter?customerType=Premium

Output:

```
[
  {
    "orderId": 5001,
    "customerName": "Arun",
    "customerType": "Premium"
  }
Etc…customers(25)
```

Q4. Revenue Summary

GET /orders/revenue/summary

Output:

```
{
  "totalRevenue": 938550,
  "cancelledOrdersCount": 12
}
```

Q5. Highest Value Order

GET /orders/highest-value

Output:

```
{
  "orderId": 5001,
  "customerName": "Arun",
  "totalAmount": 75500
}
```

SET E – Course Enrollment

Base Route: /enrollments

Q1: Highest Attendance Student

GET /enrollments/top-attendance

Output

```
{
  "id": 350,
  "studentName": "Girish",
  "course": "Machine Learning",
  "attendance": 99
}
```

Q2: Get Enrollment by ID

GET /enrollments/:id

Example

GET /enrollments/308

Output

```
{
  "id": 308,
  "studentName": "Sneha",
  "department": "CSE",
  "course": "Machine Learning",
  "instructor": "Dr. Ravi",
  "credits": 4,
  "semester": 6,
  "grade": "A",
  "attendance": 97
}
```

Q3: Filter by Department

GET /enrollments/filter?department=VALUE

Example

GET /enrollments/filter?department=CSE

Output (sample entries)

```
[
  {
    "id": 301,
    "studentName": "Arun",
    "department": "CSE",
```

```
    "course": "Data Structures",
    "instructor": "Dr. Ravi",
    "credits": 4,
    "semester": 6,
    "grade": "A",
    "attendance": 92
  },
Etc., (22 enrollments)
```

Q4: Most Enrolled Course

GET /enrollments/course/popular

Computed from dataset

Machine Learning → 7 enrollments

Output

```
{
  "course": "Database Systems",
  "totalEnrollments": 7
}
```

Q5: Attendance Analysis

GET /enrollments/attendance/summary

Computed from dataset

Average attendance → 91.48

Students above 90 → 32

Output

```
{
  "averageAttendance": 91.42,
  "above90Count": 32
}
```

SET F – Academic (Base Route: /students)

Q1: Highest Total Marks

GET /students/topper

Output:

```
{
  "id": 37,
  "name": "Student37",
  "gender": "Female",
  "totalMarks": 285
}
```

Q2: Get Student by ID

GET /students/:id

Example:

GET /students/2

Output:

```
{
  "id": 2,
  "name": "Student2",
  "gender": "Female",
  "marks": {
    "dsa": 92,
    "mern": 95,
    "aptitude": 89
  },
  "attendance": 96
}
```

Q3: Filter by MERN Marks

GET /students/filter?minMern=VALUE

Example:

GET /students/filter?minMern=98

Output:

```
[
  {
    "id": 37,
    "name": "Student37",
    "gender": "Female",
    "marks": {
```

```
      "dsa": 95,
      "mern": 98,
      "aptitude": 92
    },
    "attendance": 100
  }
]
```

Q4Gender-wise Top Scorer

GET /students/top/:gender

Example:

GET /students/top/male

Output:

```
{
  "gender": "Male",
  "id": 6,
  "name": "Student6",
  "totalMarks": 239
}
```

Q5. Attendance Analysis

GET /students/attendance/average

Output:

```
{
  "averageAttendance": 86.84,
  "above90Count": 16
}
```